

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

# Grade 6 Mathematics

**Term 3 — Lesson Plans (Lesson Exemplars)**

SY 2026–2027 · MATATAG

Plane Figures, Circles, Composite Figures, and Data & Probability  
4 A's format · aligned to the official Budget of Work

# Lesson Plans

Six lesson exemplars covering Term 3 competencies

Procedure follows Activity → Analysis → Abstraction → Application.

Lesson 1 · Properties of Triangles and Quadrilaterals MG 6MG-IIIa-1

|                |                                                              |
|----------------|--------------------------------------------------------------|
| Subject Matter | Classifying triangles and quadrilaterals by their properties |
| Time           | 2 sessions (100 min)                                         |
| Materials      | cut-out polygons, geoboards, ruler, protractor               |

LEARNING OBJECTIVES

(K) name triangles by sides/angles and quadrilaterals by their properties; (S) classify and describe each figure; (V) value precision and order.

**ACTIVITY** — “Shape Sorting.” Groups sort a pile of cut-out shapes into triangles and quadrilaterals, then into sub-groups.

**ANALYSIS** — What makes a triangle equilateral, isosceles, scalene? Right, acute, obtuse? How is a square different from a rhombus or a rectangle?

**ABSTRACTION** — **Triangles by sides:** equilateral, isosceles, scalene; **by angles:** acute, right, obtuse. The three angles sum to 180°. **Quadrilaterals:** parallelogram, rectangle, square, rhombus, trapezoid — described by parallel sides, equal sides, and right angles. Angle sum = 360°.

**APPLICATION** — Given two angles of a triangle (50° and 60°), find the third (70°) and classify it (acute).

FORMATIVE ASSESSMENT / ASSIGNMENT

Classify-5-figures slip. Assignment: find and name 3 quadrilaterals at home.

## Lesson 2 • Perimeter and Area of Plane Figures MG 6MG-IIIb-2

|                |                                                                 |
|----------------|-----------------------------------------------------------------|
| Subject Matter | Perimeter and area of triangles, parallelograms, and trapezoids |
| Time           | 2 sessions (100 min)                                            |
| Materials      | grid paper, ruler, cut-out figures, tape measure                |

### LEARNING OBJECTIVES

**(K)** state the area formulas for triangle, parallelogram, trapezoid; **(S)** compute perimeter and area; **(V)** value usefulness of measurement in daily life.

**ACTIVITY** — “*Bayong Garden.*” Groups measure a triangular and a trapezoidal “garden plot” on grid paper and count squares to estimate area.

**ANALYSIS** — How does a triangle relate to a rectangle of the same base and height? How can a parallelogram be reshaped into a rectangle?

**ABSTRACTION** — **Perimeter** = sum of all sides. **Area:** triangle =  $\frac{1}{2} \times b \times h$ ; parallelogram =  $b \times h$ ;  
trapezoid =  $\frac{1}{2} \times (b_1 + b_2) \times h$ .

**APPLICATION** — “A triangular sail has base 6 m and height 4 m. Find its area.” ( $A = \frac{1}{2} \times 6 \times 4 = 12 \text{ m}^2$ .)

### FORMATIVE ASSESSMENT / ASSIGNMENT

3 area items. Assignment: find the area of a triangular or trapezoidal object at home.

### Lesson 3 · Circles — Parts and Circumference

MG

6MG-IIIc-3

|                |                                                         |
|----------------|---------------------------------------------------------|
| Subject Matter | Identifying parts of a circle; finding circumference    |
| Time           | 2 sessions (100 min)                                    |
| Materials      | circular objects (cans, plates), string, ruler, compass |

#### LEARNING OBJECTIVES

(K) name the parts of a circle and state  $C = \pi d = 2\pi r$ ; (S) measure and compute circumference; (V) appreciate circles in nature and design.

**ACTIVITY** — “String Around.” Learners wrap string around cans/plates, measure the distance around and across, and compute the ratio ( $\approx 3.14$ ).

**ANALYSIS** — What is the relationship between the distance around (circumference) and across (diameter)? Why is it always about 3.14?

**ABSTRACTION** — **Parts:** center, radius, diameter ( $d = 2r$ ), chord, arc, circumference.  $\pi \approx 3.14$ .  
**Circumference**  $C = \pi d = 2\pi r$ .

**APPLICATION** — “A bicycle wheel has diameter 60 cm. Find its circumference.” ( $C = 3.14 \times 60 = 188.4$  cm.)

#### FORMATIVE ASSESSMENT / ASSIGNMENT

Label-the-parts diagram + 2 C items. Assignment: measure a round object’s circumference with string.

## Lesson 4 • Area of a Circle and Composite Figures MG 6MG-IIIId-4

|                |                                                             |
|----------------|-------------------------------------------------------------|
| Subject Matter | Area of a circle; area of composite figures                 |
| Time           | 2 sessions (100 min)                                        |
| Materials      | circle cut into sectors, grid paper, composite-figure cards |

### LEARNING OBJECTIVES

**(K)** state  $A = \pi r^2$ ; **(S)** find the area of a circle and of composite figures; **(V)** value careful, step-by-step problem solving.

**ACTIVITY** — “*Pizza Rearrange*.” A paper circle cut into sectors is rearranged into a near-rectangle to motivate  $A = \pi r^2$ .

**ANALYSIS** — How does the rearranged shape resemble a rectangle with base  $\approx$  half the circumference and height = radius? How do we find the area of a figure made of a rectangle plus a half-circle?

**ABSTRACTION** — **Area of a circle**  $A = \pi r^2$ . **Composite figures:** split into basic shapes, find each area, then add (or subtract for holes).

**APPLICATION** — “*Find the area of a circle with radius 7 cm.*” ( $A = 3.14 \times 7 \times 7 = 153.86 \text{ cm}^2$ .)

### FORMATIVE ASSESSMENT / ASSIGNMENT

2 circle + 1 composite item. Assignment: design a composite logo and compute its area.

## Lesson 5 • Data, Pie Graphs, and Measures of Central Tendency DP 6DP-IIIe-5 /

6DP-III f-6

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| <b>Subject Matter</b> | Constructing/interpreting pie graphs; computing mean, median, and mode |
| <b>Time</b>           | 2 sessions (100 min)                                                   |
| <b>Materials</b>      | survey slips, protractor, colored pencils, chart paper                 |

### LEARNING OBJECTIVES

**(K)** read a pie graph and define mean, median, mode; **(S)** construct a pie graph and compute the three measures; **(V)** value honesty in collecting and reporting data.

**ACTIVITY** — “*Favorite Filipino Food.*” Class surveys favorite dishes (adobo, sinigang, etc.) and tallies the results.

**ANALYSIS** — How can we show parts of a whole at a glance? How many degrees represent each category? Which single number best describes “typical”?

**ABSTRACTION** — **Pie graph:** each category’s angle =  $(\text{its part} \div \text{total}) \times 360^\circ$ . **Mean** =  $\text{sum} \div \text{count}$ ; **Median** = middle value (ordered); **Mode** = most frequent value.

**APPLICATION** — Scores 7, 8, 8, 9, 10 → mean = 8.4, median = 8, mode = 8. Make a pie graph of a 40-pupil survey.

### FORMATIVE ASSESSMENT / ASSIGNMENT

Interpret a given pie graph + compute mean/median/mode of 5 numbers. Assignment: survey 5 family members and tally.

## Lesson 6 · Simple Experimental Probability DP 6DP-IIIg-7

|                |                                                           |
|----------------|-----------------------------------------------------------|
| Subject Matter | Describing and computing the probability of simple events |
| Time           | 1–2 sessions (50–100 min)                                 |
| Materials      | coins, dice, colored marbles in a bag, spinner            |

### LEARNING OBJECTIVES

**(K)** use the terms certain, likely, unlikely, impossible and state  $P(\text{event}) = \text{favorable} \div \text{total}$ ; **(S)** compute simple probabilities; **(V)** value fairness and reasoning under uncertainty.

**ACTIVITY** — “*Lucky Draw.*” Groups draw marbles from a bag (3 red, 2 blue) and record outcomes over many trials.

**ANALYSIS** — Which color came out more often? Was it certain or just likely? How do we put a number on “chance”?

**ABSTRACTION** — **Probability**  $P(E) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$ , a value from 0 (impossible) to 1 (certain). Example: a bag of 3 red + 2 blue  $\rightarrow P(\text{red}) = \frac{3}{5}$ .

**APPLICATION** — “A die is rolled. What is the probability of getting an even number?” (Favorable 2,4,6  $\rightarrow \frac{3}{6} = \frac{1}{2}$ .)

### FORMATIVE ASSESSMENT / ASSIGNMENT

3 probability items (coin, die, marbles). Assignment: list one certain, one likely, one impossible event at home.